

Introduction

In this project, we propose a Structurally Constrained Reasoning framework to address current limitations regarding reasoning being a linear sequence of unstructured tokens, we enforce a directed acyclic graph (DAG) reasoning structure at inference time. This report evaluates the effectiveness of this DAG-based prompting across three diverse models: Llama 3.2-11B, Qwen 2.5-VL-7B, and GPT-4o on Facebook Hateful Memes and PrideMM datasets, investigating whether structural constraints can improve the performance and interpretability of different shot types without the need for specific fine-tuning related to the task to see if we could do better than state-of-the-art [ExPO-HM](#).

Approach

Based on our initial proposal, we implemented a structured DAG reasoning framework for hateful meme classification using three vision-language models: Llama 3.2-11B, Qwen 2.5-VL-7B and GPT-4o. We required each model to generate an explicit DAG before performing binary classification on its input to categorize the input meme as hateful or not hateful.

The DAG structure enforces three node types: leaf nodes, internal nodes, and a root node. The goal of leaves is to extract objective information from the meme, whether directly extractable from the meme or with the help of some inferred context, such that the foundation for the context surrounding the meme as well as what is given can be laid out structurally. This specifically includes visual leaves which translate the visual details of the input meme into text, text leaves which interpret the provided text / caption of the meme, and external leaves which pull in external information from the model's knowledge to help form the complete picture regarding what the input meme is about. The goal of internal nodes is to represent intermediate logical deductions that are the result of combining base-level knowledge of the meme, provided by leaf nodes. Information represented by leaf nodes can be grouped up to form a higher level logical deduction that represents a more holistic understanding of the input meme. A singular root node is finally used to place a classification on the input meme as either hateful or not hateful. The root node's final classification is the outcome of the model combining the deductions made in the internal nodes.

We specifically ask the model to output the DAG structure that we ask it to use in its reasoning in a JSON format. This is done for several reasons, including forcing the model to utilize the DAG structure in its reasoning for determining a classification for the input meme as well as for our own validation that the model is following the structured thought process that we are attempting to experiment with. The final classification decision was extracted from the structured JSON output, with a keyword fallback in cases where the model produces an unstructured prose.

To evaluate our DAG structure’s effectiveness in enhancing model reasoning, we evaluated models under three prompting conditions: zero-shot, one-shot and few-shot. Zero-shot prompting involved asking the model to classify the input meme by following the provided DAG structure. Importantly, no actual example meme / associated DAG was given to the model, rather we wanted to observe how much the model could infer itself. The one-shot and few-shot prompting involved providing the same DAG structure and the input meme that the model was meant to classify, but also an example meme / associated DAG pairing that demonstrated the DAG reasoning in action. The ground-truth meme text was injected into all prompts through a text placeholder field. The prompt for all shot types are in this [Google Doc](#).

Performance and Results

Model	Dataset	Shot Type	F1	Precision	Recall	Accuracy
Llama	All	0-shot	0.4180451128	0.6465116279	0.3088888889	0.5765864333
Qwen	All	0-shot	0.6867119302	0.6135181976	0.7797356828	0.6530612245
GPT-4o	All	0-shot	0.606695	0.703883	0.533088	0.697749
Llama	All	1-shot	0.5617367707	0.6369230769	0.5024271845	0.6283084005
Qwen	All	1-shot	0.6651108518	0.6209150327	0.716080402	0.6591448931
GPT-4o	All	1-shot	0.661386	0.684426	0.639847	0.711149
Llama	All	few-shot	0.5026595745	0.6798561151	0.3987341772	0.6112266112
Qwen	All	few-shot	0.6431181486	0.630071599	0.6567164179	0.6499402628
GPT-4o	All	few-shot	0.62069	0.669767	0.578313	0.699145
Llama	PrideMM	0-shot	0.4512195122	0.6548672566	0.3441860465	0.589977221
Qwen	PrideMM	0-shot	0.6799204771	0.587628866	0.8066037736	0.6365688488
GPT-4o	PrideMM	0-shot	0.527473	0.712871	0.418605	0.666667
Llama	PrideMM	1-shot	0.5333333333	0.6052631579	0.4766839378	0.6148325359
Qwen	PrideMM	1-shot	0.6545454545	0.5833333333	0.7455621302	0.6443850267
GPT-4o	PrideMM	1-shot	0.639175	0.732283	0.567073	0.716216
Llama	PrideMM	few-shot	0.5522788204	0.7304964539	0.4439655172	0.6491596639
Qwen	PrideMM	few-shot	0.6497175141	0.6182795699	0.6845238095	0.6612021858
GPT-4o	PrideMM	few-shot	0.600733	0.745455	0.503067	0.70137
Llama	FB_Hateful_Memes	0-shot	0.3857566766	0.637254902	0.2765957447	0.5642105263
Qwen	FB_Hateful_Memes	0-shot	0.6931818182	0.6398601399	0.7561983471	0.6680327869
GPT-4o	FB_Hateful_Memes	0-shot	0.712195	0.695238	0.73	0.748936
Llama	FB_Hateful_Memes	1-shot	0.5867346939	0.6647398844	0.5251141553	0.6407982262
Qwen	FB_Hateful_Memes	1-shot	0.6737288136	0.6543209877	0.6943231441	0.6709401709
GPT-4o	FB_Hateful_Memes	1-shot	0.691589	0.632479	0.762887	0.702703
Llama	FB_Hateful_Memes	few-shot	0.4538258575	0.6277372263	0.3553719008	0.5740740741
Qwen	FB_Hateful_Memes	few-shot	0.6381156317	0.6394849785	0.6367521368	0.6411889597
GPT-4o	FB_Hateful_Memes	few-shot	0.649215	0.590476	0.72093	0.695455

Figure 1: Performance of All the Models

Inter-models Analysis

Qwen2.5-VL demonstrates the most consistent performance across prompting settings, with F1 scores ranging from 0.638 to 0.693 on FB Hateful Meme and 0.645 to 0.680 on PrideMM. In contrast, GPT-4o achieves the highest zero-shot performance on FB dataset with

F1=0.712, but its performance decreases under one-shot(0.692) and few-shot(0.649) settings. The similar trends can also be observed from PrideMM, where GPT-4o improves from 0.527 (0-shot) to 0.639 (1-shot) but drops again to 0.601 (few-shot). These results suggest that in-context examples do not consistently improve performance and may introduce noise.

Also, across all settings Llama always underperforms than Qwen2.5-VL and GPT-4o. This consistent performance gap is primarily due to Llama 3.2-11B is a general-purpose language model and does not natively support multimodal reasoning, whereas Qwen2.5-VL and GPT-4o are trained with vision language, enabling them to better integrate visual and textual information.

Our Results vs Baseline

The current State of the Art approach is proposed by [ExPO-HM \(Explain-then-Detect Policy Optimization for Hateful Memes\)](#), which attempts to generate rationales before predicting a label in an attempt to solve the disconnect in multimodal interpretability. Naive implementations using standard Chain-of-Thought (CoT) prompting or standard post-training actually underperform direct Supervised Fine-Tuning (SFT) baselines. The ExPO-HM framework addressed this by utilizing GRPO with curriculum learning and a Conditional Decision Entropy (CDE) reward. ExPO-HM achieves an 81.1 F1 score on the HatefulMemes binary classification task (using Qwen2.5-VL-7B), a 15% and 17% F1 improvement over the GRPO and DPO baselines, respectively.

#	Warmup	SFT				w/ GRPO-CL and CDE			
		<i>B</i> F1 ↑	<i>Attack</i> F1 ↑	<i>Target</i> F1 ↑	<i>R</i> LLM ↑	<i>B</i> F1 ↑	<i>Attack</i> F1 ↑	<i>Target</i> F1 ↑	<i>R</i> LLM ↑
1	-	65.9	44.7	64.5	5.0	73.3	69.3	72.1	5.2
2	SFT-B	74.1	58.2	69.4	4.9	73.5	66.8	70.1	5.1
3	SFT-R	72.2	51.6	63.1	5.0	79.2	72.3	73.2	5.7
4	SFT-FG	72.5	58.4	67.7	4.9	78.9	73.4	73.4	5.6
5	SFT-PM	74.3	64.6	68.8	5.0	81.1	75.6	77.2	6.2

Figure 2: Results from ExPO-HM Paper

Our approach differs from ExPO-HM in both training paradigm and reasoning structure. ExPO-HM relies on supervised fine-tuning and reinforcement learning, where reasoning is expressed as a linear sequence of tokens.

Despite the strong zero-shot performance of GPT-4o, the full pipeline in ExPO-HM achieves substantially higher performance (up to 81.1 F1). The approaches used in the paper allow the model to adapt specifically to the hateful meme detection task, improving both classification accuracy and the quality of intermediate reasoning. In contrast, our approach relies

entirely on prompting, without updating model parameters, which limits its ability to fully specialize to the dataset.

At the same time, GPT-4o in our zero-shot setting achieves $F1 = 0.712$, outperforming the 65.9 F1 no-warmup SFT baseline. This suggests that modern multimodal foundation models already encode strong general-purpose vision-language knowledge, allowing them to perform competitively even without task-specific training. In contrast, the no-warmup SFT baseline likely starts from a less specialized or less capable initialization, and without sufficient training signals, fails to fully capture the complex multimodal interactions required for hateful meme detection.

Conclusion

Our project demonstrates that enforcing a DAG-based reasoning structure significantly enhances the transparency of multimodal hate speech detection. Notably, GPT-4o achieved an F1 score of 0.712 in a zero-shot setting, outperforming the baseline in existing literature. While our prompting-only approach does not yet match the 81.1 F1 score of ExPO-HM, it provides a low-cost, interpretable alternative that exposes logical leaps often hidden in unstructured text. Requiring models to ground their claims in verifiable visual and textual leaves may be a step closer to a moderation system that is not only accurate, but auditable as well.

Related Codes

[\[Codes used to implement GPT-4o training\]](#)

[\[Colab files used to implement Qwen and Llama training\]](#)

Contributions

Group Member Name	Contribution
Amogh Rajagopal	GPT-4o training + Report writing
Jerry Wang	Qwen training + Report writing
Jinhee Chang	Llama training + Report writing
Junyi Liu	GPT-4o training + Report writing